

topic. The `django-taggit` (<https://pypi.org/project/django-taggit/>) library comes with a `Model` that represents a tag that can be applied to any object in your database. You can then query other objects with similar tags.

- *Storage*—The content in your database isn't the only data in your system. In chapter 7, you learned about users uploading files to your project. Both the upload and static file mechanisms are built on top of Django's storage API (<https://mng.bz/1Mrz>), with the default implementation using the filesystem. The `django-storages` (<https://django-storages.readthedocs.io/en/latest/>) library provides other storage API classes, allowing you to store content in places other than the filesystem. It currently supports Amazon S3, Apache Libcloud, Azure Storage, Backblaze B2, Digital Ocean, Dropbox, FTP, Google Cloud, Oracle Cloud, and SFTP.

14.4 Web tools

Django has everything you need to get a website up and going, but there are still things that could be easier. Plenty of libraries help you with other website complications, including getting new projects started, managing JavaScript files, and providing authentication through third parties, like Google and Facebook:

- *Cookiecutter site creator*—Although the `startproject` management command creates the minimum structure of a Django project, it is still the minimum structure. Most projects use template and static file directories, and all projects begin with at least one app. The `django-cookiecutter` (<https://mng.bz/Bdgq>) tool helps you start a new project, with a lot of the features you use already inserted for you. Cookiecutter supports Bootstrap 5, uses SSL by default, and allows integration with several email clients, among many other useful features. When you run the script, it asks a series of questions and outputs a new project structure for you based on the answers.
- *JavaScript and CSS compression*—As your projects get larger, they're likely to include more JavaScript and CSS. The average webpage today is significantly larger than the days of yore. Both JavaScript and CSS can be minified; you'd be surprised how much space can be gained by removing whitespace and renaming some variables. The `django-compressor` (<https://pypi.org/project/django-compressor/>) library provides template tags that surround the JavaScript in your HTML files and compresses and rewrites these and your static CSS files. Minification means smaller pages, which means faster page loads and, therefore, happier users.
- *Customizing the Django Admin*—The Django Admin is designed around the `Model` classes in your project, but sometimes, there is value in having other kinds of pages in the Django Admin. For example, you might want a report that only administrators get to see. Instead of writing a separate view outside the Django Admin, `django-custom-admin-pages` (<https://mng.bz/d6Zz>) allows you to write CBVs that appear in the Django Admin navigation.